

Constructor Overloading

Exercise 1: Employee Salary

Create a class Employee.

Variables:

String name;

double salary;

double bonus;

Constructor overloading:

Employee(String name)

Employee(String name, double salary)

Employee(String name, double salary, double bonus)

Method overloading:

double calculateSalary()

double calculateSalary(double overtime)

double calculateSalary(double overtime, double allowance)

Rules:

calculateSalary()= salary + bonus

calculateSalary(overtime)= salary + bonus + overtime

calculateSalary(overtime, allowance)= salary + bonus + overtime + allowance

Exercise 2: Rectangle

Create a class Rectangle.

Variables:

double length;

double width;

Constructor overloading:

Rectangle()

Rectangle(double side)

Rectangle(double length, double width)

Create overloaded methods named calculate():

double calculate()

double calculate(double length)

double calculate(double length, double width)

Exercise 3: Bank Account

Create a class BankAccount.

Variables:

String accountName;

String accountNumber;

double balance;

Constructor overloading:

BankAccount(String accountName)

BankAccount(String accountName, String accountNumber)

BankAccount(String accountName, String accountNumber, double balance)

Create overloaded deposit() methods:

void deposit(double amount)

void deposit(double amount, String description)

void deposit(double amount, String description, int transactionId)

Exercise 4: Product

Create a class Product.

Variables:

String name;

double price;

int quantity;

Constructor overloading:

Product(String name)

Product(String name, double price)

Product(String name, double price, int quantity)

Create overloaded calculatePrice() methods:

double calculatePrice()

double calculatePrice(int quantity)

double calculatePrice(int quantity, double discount)

Rules:

calculatePrice()= price $\times$ object quantity

calculatePrice(quantity)= price $\times$ given quantity

calculatePrice(quantity, discount)= total price after percentage discount

Exercise 5: Mobile Phone

Create a class Mobile.

Variables:

String brand;

String model;

double price;

Constructor overloading:

Mobile(String brand)

Mobile(String brand, String model)

Mobile(String brand, String model, double price)

Method overloading:

void show()

void show(String heading)

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void show(String heading, boolean showPrice)
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Exercise 6: Book

Create a class Book.

Variables:

String title;

String author;

double price;

Constructor overloading:

Book(String title)

Book(String title, String author)

Book(String title, String author, double price)

Method overloading:

void discount(double percentage)

void discount(double percentage, double extraDiscount)

void discount(int fixedAmount)

Requirements:

- First method applies percentage discount.
- Second method applies percentage + extra discount.
- Third method deducts a fixed amount.

Exercise 7: Area Calculator

Create a class Area.

Variables:

double value1;

double value2;

Constructor overloading:

Area()

Area(double value1)

Area(double value1, double value2)

Method overloading:

double calculate(int side)

double calculate(int length, int width)

double calculate(double radius)

Calculate:

- Square area
- Rectangle area
- Circle area

Exercise 8: Customer Bill

Create a class Customer.

Variables:

String name;

int customerId;

double purchaseAmount;

Constructor overloading:

Customer(String name)

Customer(String name, int customerId)

Customer(String name, int customerId, double purchaseAmount)

Create overloaded calculateBill() methods:

double calculateBill()

double calculateBill(double discount)

double calculateBill(double discount, double tax)

Exercise 9: Car Rental

Create a class Car.

Variables:

String brand;

String model;

double rentPerDay;

Constructor overloading:

Car(String brand)

Car(String brand, String model)

Car(String brand, String model, double rentPerDay)

Method overloading:

double calculateRent(int days)

double calculateRent(int days, double discount)

double calculateRent(int days, double discount, double serviceCharge)

Exercise 10: Challenge — Shopping System

Create a class ShoppingCart.

Variables:

String customerName;

double totalAmount;

int numberOfItems;

Constructor overloading:

ShoppingCart()

ShoppingCart(String customerName)

ShoppingCart(String customerName, double totalAmount)

ShoppingCart(String customerName, double totalAmount, int numberOfItems)

Create overloaded addItem() methods:

void addItem(double price)

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void addItem(double price, int quantity)
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void addItem(String productName, double price, int quantity)
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Also create:

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void displayCart()
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